

1.

```
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```

```
n = 5
for i in range(5):
    for j in range(n):
        print("*",end=" ")
    n -= 1
    print()
```

```
i = 5
while i >= 1:
    j = 1
    while j <= i:
        print("*",end=" ")
        j += 1
    i -= 1
    print()
```

i	while i >= 1	j	while j <= i	print("*")	j += 1	i -= 1
5	True	1	1 <= 5 True	*	1+1 =2	-
		2	2 <= 5 True	**	2+1 =3	-
		3	3 <= 5 True	***	3+1 =4	-
		4	4 <= 5 True	****	4+1 =5	-
		5	5 <= 5 True	*****	5+1 =6	-
		6	6 <= 5 False	-	-	5-1 =4
4	True	1	1 <= 4 True	*	1+1 =2	-
		2	2 <= 4 True	**	2+1 =3	-
		3	3 <= 4 True	***	3+1 =4	-
		4	4 <= 4 True	****	4+1 =5	-
		5	5 <= 4 False	-	-	4-1 =3

i	while i >= 1	j	while j <= i	print("*")	j += 1	i -= 1
3	True	1	1 <= 3 True	*	1+1 =2	-
		2	2 <= 3 True	**	2+1 =3	-
		3	3 <= 3 True	***	3+1 =4	-
		4	4 <= 3 False	-	-	3-1 =2
2	True	1	1 <= 2 True	*	1+1 =2	-
		2	2 <= 2 True	**	2+1 =3	-
		3	3 <= 2 False	-	-	2-1 =1
1	True	1	1 <= 1 True	*	1+1 =2	-
		2	2 <= 1 False	-	-	1-1 =0
0	False	-	-	-	-	-

2.

```
*
**
***
****
*****
```

```
i = 1
while i <= 5:
    j = 1
    while j <= i:
        print("*",end=" ")
        j += 1
    print()
    i += 1
```

i	while i <= 5	j	while j <= i	print("*")	j += 1	i += 1
1	True	1	True	*	1+1=2	-
		2	False	-	-	1+1=2
2	True	1	True	**	1+1=2	-
		2	True	***	2+1=3	-
		3	False	-	-	2+1=3

i	while i <= 5	j	while j <= i	print("")	j += 1	i += 1
3	True	1	True	*	1+1=2	-
		2	True	**	2+1=3	-
		3	True	***	3+1=4	-
		4	False	-	-	3+1=4
4	True	1	True	*	1+1=2	-
		2	True	**	2+1=3	-
		3	True	***	3+1=4	-
		4	True	****	4+1=5	-
		5	False	-	-	4+1=5
5	True	1	True	*	1+1=2	-
		2	True	**	2+1=3	-
		3	True	***	3+1=4	-
		4	True	****	4+1=5	-
		5	True	*****	5+1=6	-
		6	False	-	-	5+1=6
6	False	-	-	-	-	-

Programming Fundamentals – Nested Loops Worksheet (2026 July 9)

1. Print a Square Pattern

```
# Write a program to print a 5 x 5 square using the * symbol.
for i in range(5):
    for j in range(5):
        print("*",end=" ")
    print()
```

2. Multiplication Table

```
# Write a program to display the multiplication tables from 1 to 5.

i = 1
while i <= 5:
```

```
j = 1
while j <= 5:
    print(f"{i*j:3}",end=" ")
    j += 1
print()
i += 1
```

3. Number Grid

Write a program to print a 4 × 4 grid containing the numbers 1 to 4 on each row.

```
i = 1
while i <= 4:
    j = 1
    while j <= 4:
        print(j,end=" ")
        j += 1
    i += 1
    print()
```

4. Right Triangle Pattern

Write a program to print a right triangle using * with 5 rows.

```
i = 1
while i <= 5:
    print("") if i == 1 else print("*",end=" ")
    j = 2
    while j <= i:
        if j == i:
            print("*")
        elif i == 5 and j < 5:
            print("*",end=" ")
        else:
            print(" ",end=" ")
        j += 1
    i += 1
```

5. Classroom Seating Plan

A classroom has 6 rows and each row has 5 desks.
Display the seating arrangement as: Row 1: Desk1 Desk2 Desk3 Desk4 Desk5.

```
raw = 1
while raw <= 6:
    print(f"Raw {raw}: ",end="")
```

```
desk = 1
while desk <= 5:
    print(f"Desk {desk} ",end="")
    desk += 1
print()
raw += 1
```

6. Ice Cream Shop

```
# An ice cream shop has 4 freezers.
# Each freezer contains 6 ice cream boxes.
# Each box contains 20 ice creams.

# Calculate:
#   Total number of ice cream boxes.
#   Total number of ice creams in the shop.

count_icecream = 0
count_box = 0
freezer = 1
while freezer <= 4:
    box = 1
    while box <= 6:
        count_box += 1
        ice = 1
        while ice <= 20:
            count_icecream += 1
            ice += 1
        box += 1
    freezer += 1

print(f"Ice Cream Boxes x {count_box}")
print(f"Ice Creams x {count_icecream}")
```

7. School Examination

```
# A school has 5 classrooms.
# Each classroom has 25 students.
# Each student pays Rs. 500 as the examination fee.

# Calculate:
#   Total number of students.
#   Total examination fee collected.

count = 0
exam_fee = 0
clzroom = 5
while clzroom >= 1:
    student = 25
```

```
while student >= 1:
    count += 1
    exam_fee += 500
    student -= 1
    clzroom -= 1

print(f"Total Number of Students ={count}")
print(f"Total Examination Fee Collected = Rs {exam_fee:,.2f}")
```

8. Orange Farm

```
# An orange farm has 8 trees.
# Each tree produces 35 oranges.
# Calculate the total number of oranges harvested.

count = 0
tree = 8
while tree >= 1:
    orange = 35
    while orange >= 1:
        count += 1
        orange -= 1
    tree -= 1

print(f"Total Number of Oranges Harvested = {count}")
```

9. Hotel classrooms

A hotel has 6 floors. Each floor has 18 rooms. Each room costs Rs. 12,000 per night. Assume every room is occupied.

Calculate:

Total number of rooms.

Total income for one night.

```
count = 0
income = 0
floor = 6
while floor >= 1:
    room = 18
    while room >= 1:
        count += 1
        income += 12000
        room -= 1
    floor -= 1

print(f"Total number of rooms = {count}")
print(f"Total income for one night = Rs {income:,.2f}")
```

10. Airline Passengers

An airline operates 5 flights. Each flight has 30 rows and each row has 6 seats. Assume every seat is occupied.

Calculate:

Number of passengers on one flight.

Total number of passengers on all flights.

```
passengers = 0
flights = 5
n = flights
while flights >= 1:
    row = 30
    while row >= 1:
        seat = 6
        while seat >= 1:
            passengers += 1
            seat -= 1
        row -= 1
    flights -= 1

print(f"Number of passengers on one flight x {passengers / n:.0f}")
print(f"Total number of passengers on all flights x {passengers}")
```